

POKHARA UNIVERSITY

Level: Bachelor

Semester : Spring

Year : 2023

Programme: BE

Full Marks: 100

Course: Electronic Devices and Circuits

Pass Marks: 45

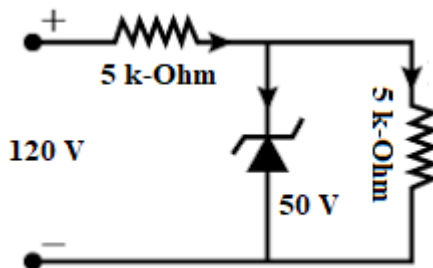
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

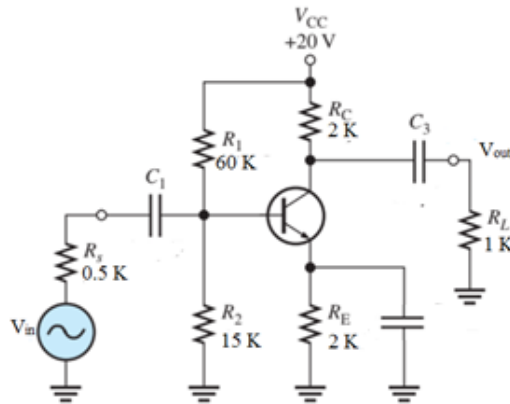
Attempt all the questions.

1. a) What are different types of semiconductors? Explain the effect of temperature in ordinary p-n junction diode. 7
- b) For the circuit shown in figure below find: 8
 - i) The output voltage
 - ii) Voltage drop across the series resistance
 - iii) Current through the Zener diode

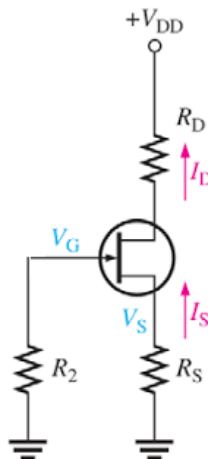


Also, Draw IV characteristic curve of Zener diode

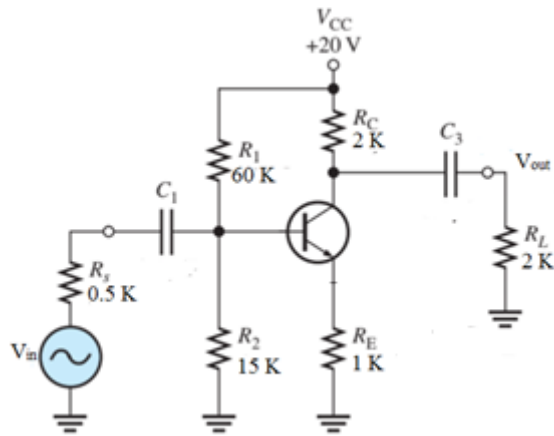
2. a) Explain the input and output characteristics of common emitter transistor with necessary diagrams and characteristic curve. 7
- b) For the BJT common emitter circuit shown below, determine the operating point and locate it in the dc load line. Assume $\beta = 100$. 8
Also find V_C and V_E



3. a) What is rectification? Explain the operation of half wave rectifier circuit with necessary circuit diagrams and waveforms. 7
- b) For the circuit given, Find out drain current and drain to source voltage at operating point. Given $I_{DSS} = 8\text{ mA}$, $V_p = -6\text{ V}$, $R_D = 3.3\text{ k}\Omega$, $R_S = 1000\text{ }\Omega$, $R_2 = 1\text{ M}\Omega$, $V_{DD} = 30\text{ V}$. Indicate value of operating point in characteristic curve 8

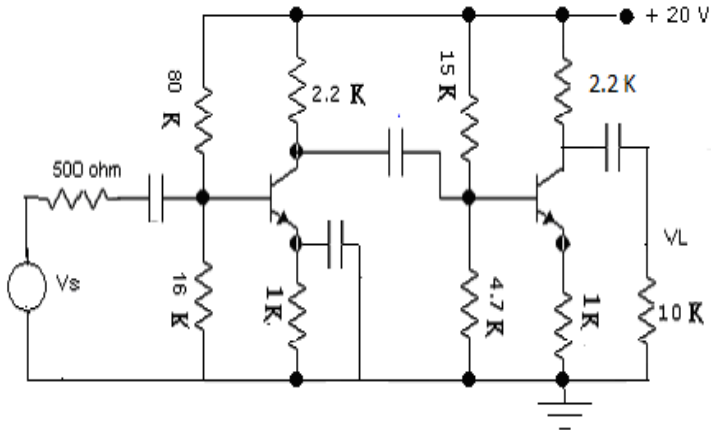


4. a) Find overall voltage gain using r_e model. Assume $\beta = 100$. 8



- b) Using r_e model, find V_L/V_S for the cascaded BJT amplifier given below. Assume $r_{e1} = r_{e2} = 20\ \Omega$, $\beta_1 = \beta_2 = 100$

7



5. a) What do you mean by power amplifier? Show that maximum possible efficiency of transformer coupled load class A power amplifier is 50 %.
- b) Negative feedback decreases the gain but increases the gain stability in a feedback amplifier. Do you agree with this statement? Justify your answer. The overall gain of a multistage amplifier is 140. When negative feedback is applied, the gain reduces to 17.5. Find the fraction of the output that is feedback to the input.
6. a) What is the importance of positive feedback on oscillator circuits? Explain RC phase shift oscillator with its necessary derivations.

7

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8

- b) Design an operational amplifier that produces an output equal to
 $V_o = -2V_1 + 8V_2 - 3V_3 + 6V_4$.

7

7. Write short notes on: (**Any two**)

2×5

- a) Regulated and unregulated power supply
- b) Small signal model of AC
- c) Concept of DC and AC analysis